The cubic density of a minimal counterexample to the Erdős–Gyárfás conjecture: a reformulation, a machine-checked strict bound, and an obstruction

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Abstract

The Erdős–Gyárfás conjecture asserts that every finite graph of minimum degree at least 3 contains a cycle whose length is a power of two. Carr proved that in a minimal counterexample at least 4/7 of the vertices are cubic; Bisch improved this to 2/3. We record three things. First, a reformulation: writing S_3 for the number of cubic-to-cubic incidences, every minimal counterexample satisfies $4|V_{\geq 4}| + S_3 \leq 3|V_3|$, so the constant in any such density bound is decided entirely by how many edges the cubic set spans among itself; Bisch's 2/3 is exactly the case in which $G[V_3]$ is a perfect matching. Second, a machine-checked proof, in Lean 4 against `Mathlib`, of the strict bound $|V_3| > \frac{2}{3}|V|$, an argument due to the forum user `jul059` and previously unverified. Third, a conditional improvement: if $G[V_3]$ has no K_2 component then $|V_3| \geq \frac{12}{17}|V|$, together with an explanation of why removing that hypothesis resists the standard minimality arguments, and an observation that no counting argument built on the known structural lemmas can beat 2/3. All statements below are formalized; nothing here is asserted that the Lean development does not prove.

1 Introduction

Conjecture 1.1 (Erdős–Gyárfás)**.** *Every finite graph with minimum degree at least* 3 *contains a simple cycle whose length is a power of two.*

The conjecture is open. Liu and Montgomery [5] proved it for graphs whose minimum degree exceeds an absolute constant, which also refuted the stronger conjecture of Erdős and Gyárfás; consequently a counterexample can only have very small minimum degree. Markström [6] showed by computer search that a cubic counterexample needs at least 30 vertices, and Garcia [3] recently established, by a SAT search certified with DRAT proofs, that any counterexample has at least 24 vertices.

Throughout, G is a *minimal counterexample*: a counterexample of minimum order and, subject to that, of minimum size. Write

$$V_3 = \{v : \deg v = 3\}, \qquad V_{\geq 4} = \{v : \deg v \geq 4\}.$$

Carr [2] proved that $V_{\geq 4}$ is an independent set, that every vertex has a neighbour of degree exactly 3, and that $|V_3| \geq \frac{4}{7}|V|$. Bisch [1] formalized those structural facts in Lean 4 and improved the density to $|V_3| \geq \frac{2}{3}|V|$.

*Formalization and write-up. The strictness argument of Section 3 is due to the Erdős Problems forum user `jul059`; see Section 7.

Our contribution is organised as follows. Section 2 gives the counting reformulation, which makes the source of the constant explicit. Section 3 reports the machine-checked strict bound. Section 4 proves the conditional 12/17 bound, and Section 5 explains why we could not remove its hypothesis — including the observation that counting alone cannot.

Standing hypotheses. All results are stated for a graph satisfying the properties of a minimal counterexample that we actually use: minimum degree at least 3; $V_{\geq 4}$ independent; every vertex has a cubic neighbour; no 4-cycle; minimality in the order; and the absence of a power-of-two cycle. In the formalization these are bundled as `MinCexHyps`, and `EGCBridge` derives that bundle from Bisch's `IsMinCex`, so every statement below also holds verbatim for a minimal counterexample in his sense.

2 The reformulation

Definition 2.1. For a vertex v let $\mathrm{cd}(v)$ denote its number of cubic neighbours, and put

$$S_3 \;=\; \sum_{v \in V_3} \mathrm{cd}(v),$$

the number of ordered cubic-to-cubic incidences; equivalently $S_3 = 2\, e(G[V_3])$.

Theorem 2.2. $4|V_{\geq 4}| + S_3 \leq 3|V_3|$. [Lean: `four_card_big_add_sum_cubicDeg_le`]

Proof. Since $V_{\geq 4}$ is independent, every neighbour of a vertex of $V_{\geq 4}$ is cubic, so $\deg u$ counts cubic neighbours for each $u \in V_{\geq 4}$ and $\sum_{u \in V_{\geq 4}} \deg u = e(V_{\geq 4}, V_3)$. Counting that quantity from the other side and using that a cubic vertex has exactly $3 - \mathrm{cd}(v)$ neighbours in $V_{\geq 4}$,

$$4|V_{\geq 4}| \;\leq\; \sum_{u \in V_{\geq 4}} \deg u \;=\; e(V_{\geq 4}, V_3) \;=\; \sum_{v \in V_3} \big(3 - \mathrm{cd}(v)\big) \;=\; 3|V_3| - S_3. \qquad \square$$

Theorem 2.2 isolates what controls the constant: a density bound follows from any lower bound on S_3, and from nothing else.

Corollary 2.3 (Bisch's bound)**.** *$4|V_{\geq 4}| \leq 2|V_3|$, hence $|V_3| \geq \frac{2}{3}|V|$.* [Lean: `card_big_le_of_domination`]

Proof. Every cubic vertex has a cubic neighbour, so $\mathrm{cd}(v) \geq 1$ on V_3 and $S_3 \geq |V_3|$. Substitute into Theorem 2.2. $\square$

Observation 2.4. Equality $S_3 = |V_3|$ holds precisely when $G[V_3]$ is a perfect matching. So 2/3 is exactly the bound obtained in that case, and a K_2 component of $G[V_3]$ — a pair of adjacent cubic vertices whose four remaining edges all leave to $V_{\geq 4}$ — is the unique local configuration attaining the extremal ratio.

3 The strict bound, verified

Theorem 3.1. *$|V_3| \geq 2|V_{\geq 4}| + 1$, hence $3|V_3| > 2|V|$.* [Lean: `strict_two_thirds`]

The argument is not ours; it is due to `jul059` [4], who posted it on the Erdős Problems forum marked as unverified. We state it for completeness and because the formalization is what we contribute.

Suppose $|V_3| = 2|V_{\geq 4}|$. Then every inequality in the proof of Theorem 2.2 is tight, which forces every vertex of $V_{\geq 4}$ to have degree exactly 4 and every cubic vertex to have exactly two neighbours in $V_{\geq 4}$ (`degree_eq_four_of_equality`, `card_big_nbrs_eq_two_of_equality`). Contract each cubic vertex to an edge joining its two $V_{\geq 4}$ neighbours. The result H is a graph on $V_{\geq 4}$; it is simple because two cubic vertices with the same pair of $V_{\geq 4}$ neighbours would close a 4-cycle, and it is 4-regular because the four cubic neighbours of a vertex $u \in V_{\geq 4}$ lead to four *distinct* vertices — a collision would again close a 4-cycle through u (`contract_degree_ge`). Since $|V(H)| = |V_{\geq 4}| < |V|$ and H has minimum degree 4, minimality supplies a cycle in H of length 2^k. Re-inserting the cubic vertices turns it into a cycle of length 2^{k+1} in G, contradicting the hypothesis on G.

Remark 3.2. The lifting step is where the formalization does work the informal argument leaves implicit: one must check that the resulting closed walk is a *cycle*. The $V_{\geq 4}$ vertices are distinct because the cycle in H is one; the inserted cubic vertices are distinct because, in the equality case, a cubic vertex has exactly two $V_{\geq 4}$ neighbours and therefore *determines* the contraction edge it came from (`mid_determines_pair`), and a cycle traverses each edge once; and the two families are disjoint because a cubic vertex has degree 3 while a $V_{\geq 4}$ vertex does not. See `exists_pow2_cycle_of_contract_cycle`.

4 A conditional improvement

By Observation 2.4, improving on 2/3 means bounding S_3 away from $|V_3|$, i.e. excluding K_2 components of $G[V_3]$.

Definition 4.1. G has *no K_2 component* if every cubic vertex with exactly one cubic neighbour has that neighbour possess at least two. `noK2Component`

Lemma 4.2. *If G has no K_2 component then* $3S_3 \geq 4|V_3|$. [Lean: `three_mul_sum_cubicDeg_ge`]

Proof. Split V_3 into $T = \{v : \mathrm{cd}(v) = 1\}$ and $S = \{v : \mathrm{cd}(v) \geq 2\}$; domination excludes $\mathrm{cd}(v) = 0$, so this is a partition and $S_3 = |T| + \sum_{s \in S} \mathrm{cd}(s)$. Write $D = \sum_{s \in S} \mathrm{cd}(s)$, so $D \geq 2|S|$. The hypothesis sends each $v \in T$ to a cubic neighbour lying in S; the fibre of $s \in S$ consists of cubic neighbours of s, so it has at most $\mathrm{cd}(s)$ elements and hence $|T| \leq D$. Now

$$S_3 = |T| + D \geq |T| + \max(2|S|,\, |T|),$$

and comparing the two cases $|T| \geq 2|S|$ and $|T| < 2|S|$ gives $S_3 \geq \frac{4}{3}(|T| + |S|) = \frac{4}{3}|V_3|$ in both. □

Note that the proof is a fibre count; no connectivity or spanning-tree argument is needed.

Theorem 4.3. *If G has no K_2 component then* $12|V| \leq 17|V_3|$, *i.e.*

$$|V_3| \ \geq\ \tfrac{12}{17}|V| \ \approx\ 0.7059\,|V|.$$

[Lean: `twelve_seventeenths_of_noK2`]

Proof. Combine Theorem 2.2 with Lemma 4.2: $4|V_{\geq 4}| \leq 3|V_3| - \frac{4}{3}|V_3| = \frac{5}{3}|V_3|$, so $12|V_{\geq 4}| \leq 5|V_3|$. Adding $12|V_3|$ to both sides and using $|V| = |V_3| + |V_{\geq 4}|$ gives the claim. □

5 Why the hypothesis resists removal

Removing the hypothesis of Theorem 4.3 would give 12/17 outright. We did not manage it, and we record two obstacles, both of which we checked, so that the route is not re-attempted blindly.

Local replacements shift cycle lengths by one

Bisch [1] constrains the offending configuration tightly: if v, v' are adjacent and every other neighbour of either has degree at least 4, then v and v' have a *unique* common neighbour u, and $\deg u = 4$. So each K_2 component comes with a triangle $vv'u$ whose apex has degree exactly 4, its two remaining neighbours being cubic.

Every way we found of destroying such a component changes cycle lengths additively:

- delete v, v' and join u to the two outer $V_{\geq 4}$ neighbours: a 2^k cycle upstairs lifts to $2^k + 1$ or $2^k + 2$;
- delete v, v', u and join the apex's other two neighbours: lifts to $2^k + 1$;
- contract the triangle to a single vertex: shortens cycles by 0 or 1 depending on the entry and exit edges, hence non-uniformly;
- delete the apex and rewire its neighbours in pairs: lifts to $2^k + 1$.

Since $2^k + 1$ and $2^k + 2$ are not powers of two, minimality yields no contradiction in any of these cases. The obstruction is structural rather than a failure of ingenuity: replacing a 2-path by an edge shifts every cycle through it by exactly one, and the powers of two are too sparse for an additive shift to land on another one. The argument of Section 3 escapes this only because in the perfect-matching case the operation is *uniformly multiplicative* — G is a subdivision of the contraction, every cycle doubles, and $2^k \mapsto 2^{k+1}$. A single edge inside V_3 destroys that uniformity.

Counting alone cannot beat two thirds

More is true: the extremal configuration appears to be consistent with all of the structural facts currently available. Take $G[V_3]$ to be a perfect matching in which every degree-4 apex serves two K_2 components — so $u \sim v, v', p, p'$ with $v \sim v'$ and $p \sim p'$ — and let the remaining $V_{\geq 4}$ vertices have degree 4, each absorbing one edge from four distinct components. Then $V_{\geq 4}$ is independent, every vertex has a cubic neighbour, the uniqueness in Bisch's proposition is respected, no 4-cycle is forced, and the count closes at exactly $|V_{\geq 4}| = |V_3|/2$.

Consequently no argument that uses only Theorem 2.2 together with the known structural lemmas can improve on 2/3: any improvement must invoke the cycle condition, which is precisely what the additive obstruction above blocks. Breaking the deadlock needs either an operation on the K_2 configuration that scales all cycle lengths multiplicatively, or a new structural lemma about the apexes.

6 Formalization

Every numbered statement above is proved in Lean 4 against `Mathlib`, with no use of `sorry`. The development is organised as

File	Contents
`EGCStrict.lean`	hypothesis bundle, counting, the equality analysis, the contraction
`EGCLift.lean`	the cycle lifting; Theorem 3.1
`EGCDensity.lean`	Theorem 2.2, Corollary 2.3, Theorem 4.3
`EGCBridge.lean`	derives the hypothesis bundle from Bisch's `IsMinCex`

Each theorem was checked with `#print axioms` to depend only on `propext`, `Classical.choice` and `Quot.sound`; the audit files are `AxiomCheck.lean`, `BridgeCheck.lean` and `DensityCheck.lean`. Bisch's file is not redistributed — its repository carries no licence — so `EGCBridge` is not a default build target and expects the file to be fetched locally; the other three modules build and prove their statements without it, conditional on the hypothesis bundle.

7 Attribution

The mathematics of Section 3 is due to the forum user `jul059` [4]. Their comment supplies the equality analysis, the contraction, the reason it is simple, and the doubling of cycle lengths; it was posted explicitly as unverified. Our contribution there is the formalization, including the distinctness bookkeeping. The structural lemmas are Carr's [2], formalized by Bisch [1], who also proved the non-strict 2/3 bound. Sections 2, 4 and 5 are ours.

We would welcome direction from `jul059` on how they wish to be credited; we have used the handle because no name is attached to the posting.

References

[1] A. Bisch. *On the density of cubic vertices in a minimal counterexample to the Erdős–Gyárfás conjecture*. Zenodo, DOI: `10.5281/zenodo.21574476`. Lean formalization: `https://github.com/AJBisch/AJBisch.github.io/blob/main/EGC.lean`.

[2] A. Carr. *Every minimal counterexample to the Erdős–Gyárfás conjecture is predominantly cubic.* arXiv:2605.22844.

[3] D. Garcia. *Small graphs without power-of-two cycles: a lower bound of 24, a correction to a construction of Exoo, and explicit bounds for $f(k)$.* arXiv:2609.04686.

[4] `jul059`. Comment on Erdős Problem #64, 26 July 2026. `https://www.erdosproblems.com/forum/thread/64#post-8130`.

[5] H. Liu and R. Montgomery. *A solution to Erdős and Hajnal's odd cycle problem*, and related work on cycle lengths in graphs of large average degree. See the discussion at `https://www.erdosproblems.com/64`.

[6] K. Markström. *Extremal graphs for some problems on cycles in graphs.* Congressus Numerantium 171 (2004).